Exercise 5: Threshold Logic Unit

Task 1:

Given the following Threshold Logic Unit and its (usually randomly but here statically) initialized weights, complete the table below, describing the training procedure for the Boolean identity function, i.e., $y = x$. The TLU is initialized with $w = 0$, $\theta = 1.0$, and uses the learning rate $\eta = 0.3$. *You can compute these values with the help of your TLU implementation from Task 3 but try to calculate at least one step per hand.*

iteration	*step*	x	o	xw	y	$o - y$	$\Delta\theta$	Δw	θ	w
...										

Task 2:

Test graphically whether you can find a linear function that acts as a perfect classifier for the Boolean **XOR** function.

Task 3:

Implement a TLU using Python 3. It should be able to approximate these basic Boolean functions:

- *Not*
- *Identity*
- *And*
- *Or*
- *Implication*
- *Bi-Implication*

The TLU is initialized with learning rate $\eta = 0.3$, weights $\forall w_i \in w,\ w_i = 0$ and threshold $\theta = 1.0$. Complete the following table for all Boolean functions mentioned above.

Boolean function	*Number of steps*	e after termination
...		

How does the training output change when changing weight initialization or learning rate?

Task 4:

Try to apply the TLU you implemented in Task 3 to the survival prediction task from the previous exercise sheet. Use the same features as you did in the previous exercise sheet.

How does the TLU compare to the decision tree and the guessing model?
Use the same metrics as in the previous exercise sheet.